

# MD SHAKHRUL IMAN SIAM

Columbus, Ohio, USA

✉ siam.5@osu.edu | ☎ +1 (614) 205 9451

in shakhrul-iman-siam | 🌐 shakhrulsiam268 | 🌐 <https://shakhrulsiam268.github.io/>

## RESEARCH INTEREST

My research centers on **multimodal LLMs**: building multimodal foundation models and agentic systems that perceive long-horizon, real-world signals from egocentric video, gaze, and audio, and reason over them efficiently.

**Keywords:** MLLMs, VLMs, Egocentric Long Video Understanding, Long-Context and Memory, Agentic Workflows, Audio-Language Models, Automatic Speech Recognition (ASR), Efficient Multimodal Inference.

## WORK EXPERIENCE

### Microsoft

Redmond, WA, USA

- Applied Science Intern

May 2026 - August 2026

#### Research Focus: LLM, Agentic workflow

##### ◦ LLM Agent for PowerPoint Slide Generation

- Built task-complexity-based model routing for PowerPoint Agent, reducing inference COGS by routing simple tasks to fine-tuned open-source models with frontier LLM fallback.
- Replace VLM-based visual QA with deterministic Python scripts for overlap detection, off-slide-edge detection, and layout operations.
- Reduced 21% average token reduction and 16% latency reduction by eliminating redundant Visual pass without affecting performance.

### Microsoft

Remote, USA

- Research Intern

June 2025 - August 2025

#### Research Focus: ASR, Voice Privacy, Neural Audio Codec

##### ◦ Neural Audio Codec for Secured Voice Transmission

- Designed a Neural Audio Codec that disentangles speaker identity from voice data and transmits with a very low bitrate.
- Achieved 13.5% improvement of Equal Error Rate compared to state-of-the-art Voice conversion system with only 6.6% degradation of relative WER compared to original audio.
- Fine-tune an LLM-based Automatic Speech Recognition System that uses the transmitted codes directly as input without the need to reconstruct to the audio waveform.

### The Ohio State University

Columbus, Ohio, USA

- Graduate Research Associate, OSU AIoT and Machine Learning Systems Lab 

August 2023 - Present

#### Advisor: Prof. Mi Zhang

##### ◦ Multimodal Audio-language Foundation Model for Respiratory Health Diagnosis

ACL 2026

- Developed a Multimodal Foundation model that will capture sounds from a smartphone microphone and perform several downstream tasks like disease (Covid-19, TB, COPD) detection, generating health reports, and Question-answering.
- Utilize Large Language Models with PEFT and LoRA techniques to fine-tune multimodal models from cough sounds and self-reported patient text metadata.

##### ◦ A large-scale multi-modal reading dataset of 100 hours of egocentric videos

NeurIPS 2025

- A first-of-its-kind dataset that enables egocentric and spatial AI on the next-generation Meta AI glasses.
- Developed a transformer-based model that detects whether and when a user is reading using different input modalities (e.g., egocentric video, eye gaze, IMU)
- Developed a benchmark using both gaze information and RGB image data to classify reading and non-reading activities.

##### ◦ Gaze-guided Keyframe Selection for Egocentric Long Video Understanding

- Developed a training-free keyframe selection framework for long egocentric video understanding, achieving up to 15.7% accuracy improvement over SOTA sampling methods across various Vision Language Models(VLMs).
- Designed a cross-modal query-frame retrieval pipeline and cost-aware Maximal Marginal Relevance (MMR) selection for keyframe selection.

##### ◦ Visually-Grounded Entity Graph for Egocentric Long Video Question Answering

- Designed VEG-QA, a training-free multi-agent system for long-form egocentric video QA that unifies visual embeddings and structured graph representations.
- Constructed a visually-grounded entity graph over week-long egocentric streams with typed nodes (episodes, persons, objects, locations, activities) storing CLIP embeddings alongside textual attributes.
- Outperformed SOTA agentic baselines on EgoLife QA by replacing disconnected visual stores with unified graph-native visual grounding.

- ACI Limited** Dhaka, Bangladesh  
 • Machine Learning Engineer October 2021 - July 2023  
**Domain Experience: Computer Vision, Speech Recognition, Predictive analysis, NLP**
- **Person re-identification and tracking using multiple CCTV camera feeds.** 
    - Developed a system to re-identify, track, and monitor customers' movement in Retail outlets from CCTV camera.
    - Person Detection using YOLOv5 model and Re-identification using embedding generated by a Resnet-50 model.
  - **Facial Recognition based automatic attendance system.** 
    - Implemented face detection using Retinaface and face recognition using Arcface.
    - Converted the model to Tflite to improve inference speed by 4×.
  - **Medicine name recognition system from handwritten prescription.** 
    - Developed a ROI Extraction model by using LayoutLM.
    - Developed OCR models using EasyOCR, PyTesseract, and PaddleOCR to recognize medicine names.
  - **Voice Controlled Conversational Chatbot.** 
    - Implemented a streaming ASR model and a Text-to-speech model for automated speech recognition of users.
    - Designed a sliding window technique to provide the ASR model with real-time streaming capabilities.

## EDUCATION

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- **The Ohio State University** Columbus, Ohio, USA  
 Ph.D. in Computer Science and Engineering (CSE) 08/2023 - 12/2027 (Expected)
- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh  
 BSc. in Electrical and Electronic Engineering (EEE) 02/2016 - 02/2021

## PUBLICATIONS

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### Computer Vision

- [MemoLens: Empowering Augmented Reality Glasses with Super Memory](#)  
 Samiul Alam, *Shakhrul Iman Siam*, Mi Zhang  
**Mobisys 2026**
- [Reading Recognition in the Wild](#)  
 Charig Yang, Samiul Alam, *Shakhrul Iman Siam*, Michael J. Proulx, Lambert Mathias, Kiran Somasundaram, James Fort, Omkar Parkhi, Yuheng Ren, Mi Zhang, Yuning Chai, Richard Newcombe, Hyo Jin Kim  
**NeurIPS 2025**
- [SuperMemory-VQA: An Egocentric Visual Question Answering Benchmark for Long-Horizon Memory](#)  
 Samiul Alam, *Shakhrul Iman Siam*, Michael J. Proulx, James Fort, Richard Newcombe, Hyo Jin Kim, Mi Zhang  
**Under Review**
- [GazeFrame: Gaze-guided Keyframe Selection for Egocentric Long Video Understanding](#)  
*Shakhrul Iman Siam*, Samiul Alam, Mi Zhang  
**Under Review**
- [A Deep Learning-Based Person Detection and Heatmap Generation Technique with a Multi-Camera System](#)  
*Md Shakhrul Iman Siam*, Subrata Biswas  
**ICECE 2022**
- [A Dual-Purpose Refreshable Braille Display Based on Real-Time Object Detection and Optical Character Recognition](#)  
 KM Naimul Hassan, Subrata Kumar Biswas, Md Shakil Anwar, *Md Shakhrul Iman Siam*, Celia Shahnaz  
**SPICSCON 2019**

### Audio, Speech and NLP

- [RespiraMFM: A Multimodal Foundation Model with Contrastive Audio-Language Alignment for Respiratory Disease Identification](#)  
*Shakhrul Iman Siam*, Tiantian Feng, Jiankun Zhang, Shrikanth Narayanan, Mi Zhang  
**ACL 2026 (Main)**
- [ChildVox: A Speech, Audio, and Large Audio-Language Model Benchmark in Understanding and Characterizing Sound across Childhood](#)  
 Tiantian Feng, Anfeng Xu, Xuan Shi, Aditya Kommineni, *Shakhrul Iman Siam*, Megan Micheletti, Zhonghao Shi, Helen Tager-Flusberg, Mi Zhang  
**EMNLP 2026 (Main)**

- [DECAF: A Privacy Preserving Speech Codec Using Speaker Disentanglement and Canonical Voice Conversion](#)  
*Shakhrul Iman Siam*, Dushyant Sharma, Stanislav Yu. Kruchinin, Peter Skala  
**IWAENC 2026**

## Signal Processing

- [Artificial Intelligence of Things: A Survey](#)  
*Shakhrul Iman Siam*, Hyunho Ahn, Li Liu, Samiul Alam, Hui Shen, Zhichao Cao, Ness Shroff, Bhaskar Krishnamachari, Mani Srivastava, and Mi Zhang.  
**TOSN 2024**
- [ChirpTransformer: Versatile LoRa Encoding for Low-power Wide-area IoT](#)  
Chenning Li, Yidong Ren, Shuai Tong, *Shakhrul Iman Siam*, Mi Zhang, Jiliang Wang, Yunhao Liu, and Zhichao Cao.  
**Mobisys 2024**
- [Bioradiolocation-Based Multi-Class Sleep Stage Classification Using Time and Frequency Features with Random Forest Classifier](#)  
*Md Shakhrul Iman Siam*, Md Saiful Bari Siddiqui, Mushfiqul Abedin, and Mohammed Imamul Hassan Bhuiyan.  
**ICECE 2022**

## SKILLS SUMMARY

|                                   |                                                                                                                         |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>                | Python, C/C++, CUDA, Bash/Shell, SQL, MATLAB                                                                            |
| <b>ML Frameworks</b>              | PyTorch, HuggingFace (Transformers, Datasets, Accelerate), PEFT/LoRA, TRL, DeepSpeed, FSDP, scikit-learn, NumPy, Pandas |
| <b>LLM &amp; Multimodal</b>       | SFT, LoRA/QLoRA, retrieval-augmented generation (RAG), prompt/context engineering                                       |
| <b>Agentic Systems</b>            | Tool use and function calling, multi-agent orchestration, model routing, MCP, agent evaluation harnesses                |
| <b>Inference &amp; Efficiency</b> | vLLM, SGLang, quantization, KV-cache optimization, ONNX, TensorRT-LLM, MLC-LLM, latency and cost profiling              |
| <b>Training Infrastructure</b>    | Multi-node distributed training, Slurm, Ray, Docker, Kubernetes, AWS, Azure, Git                                        |
| <b>Evaluation &amp; Data</b>      | Benchmark design, LLM-as-judge, human evaluation protocols, large-scale data curation                                   |
| <b>Hardware &amp; Sensing</b>     | Meta Aria Glasses, eye tracking, IMU, Raspberry Pi                                                                      |

## SELECTED PROJECTS

- **Generating Customer Heatmap of a retail outlet from CCTV camera :** Using Yolov5 object detection model, Homography transformation and Kernel density estimator to generate customer density map on a floorplan of a retail outlet using video taken from CCTV camera. [↗](#)
- **Blind vision Assistance :** Prototype of a low-cost mobile device, that can assist a visually disabled person to read, recognize person, and detect objects using voice command. There are three main features of this system- Object detection, Face Recognition, and optical character recognition. [↗](#)
- **Real-time vehicle detection and tracking at junction using a fisheye camera :** The Video and Image Processing cup 2020 challenge focuses on fisheye cameras mounted into street lamps at junctions, vehicle detection and tracking to be used for a junction management system. [↗](#)
- **Open Source Contribution in developing a pypi package:** A Python library for offline reverse geocoding. Reverse Geocode BD takes a (latitude , longitude) coordinate and returns the Division, District, Upazila and Thana of any location in Bangladesh. [↗](#)
- **Unsupervised Synthetic/Fake Speech Detection.** Our research introduces a novel unsupervised model, which we have termed "Anocoder". This model employs a one-dimensional convolutional autoencoder and is designed to differentiate between authentic and synthetic or fabricated speech.
- **Web Scrapping and Sentiment Analysis from Social Media post:** Using Selenium for web scrapping and analysis of social media and online newspaper posts for sentiment analysis. Tech: Selenium, BeautifulSoup

## HONORS AND AWARDS

- First Runner up at Robi **Datathon 2.0** (2022)
- 5th at **IEEE Video and Image Processing Cup** (2020)
- Champion of Bangladesh Section and World Finalist, **IEEE YESIST12 Innovation Challenge** (2019)
- 1st Runner up, BUET CSE Day **Math Olympiad** (2017)
- 1st Runner up, BAS Divisional **Science Olympiad** (2015)
- Champion , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2015)
- 2nd Runner Up , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2014)
- Champion , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2011)